

GEO-INFORMATICS (MN542)

OBJECTIVE: Objective of the course is to disseminate the knowledge of the Geo-informatics to the students. Students will understand the fundamental of the subject and its application in mining and other industry.

OUTCOME OF THE COURSE: Students will gain the knowledge of the Geo-informatics. They will be acquainted with the different tools of geo-informatics like Remote Sensing, Geological Information System, and GPS. This will enable them to plan a mine in a more economical way with less environmental degradation.

Syllabus

Introduction to Geo-informatics and its Application to Mining Engineering.

Principles of Geo-informatics

Basic concepts.

Remote Sensing

Introduction, Remote sensing system; Physics of remote sensing, EMR characteristics and interaction in atmosphere and with ground objects, Spectral reflectance of vegetation, soil, and water; Types of Platforms, Sensor types & characteristics, Resolutions: Spatial, Spectral, Radiometric and temporal resolution, Remote Sensing satellites, Multispectral Remote sensing, Hyperspectral Remote Sensing, Thermal Remote sensing, Visual image interpretation; Microwave Remote Sensing; Applications of Remote Sensing.

Geographic Information System (GIS)

Introduction; Component of GIS, GIS data: spatial and non-spatial, spatial data model: raster, vector, Topology and topological models; Database; GIS analysis functions: Retrieval, classification, measurement, neighbourhood, topographic, interpolation, overlay, buffering, spatial join and query, connectivity, other functions; Applications.

Global Positioning System (GPS)

Introduction to geodesy, Earth and its size and shape, earth and its motions; Introduction to GNSS, Global Positioning System (GPS), Satellite constellation, GPS signals and data; GPS coordinate systems: WGS-84; GPS Errors and biases; GPS receiver, GPS antenna. Other satellite based navigational systems: GLONASS, GALILEO, Beidou, Surveying with GPS; Planning and field observations, GPS observation methods; GPS Applications.